

ENTERPRISE AI INFRASTRUCTURE

Fictional Northstar Private RAG TCO

Fictional demonstration comparing CPU inference estate and GPU inference estate.

Fictional demonstration. Illustrative assumptions are not live vendor quotes, and modeled ROI is not a guarantee.

Executive summary

Decision posture

VALIDATION REQUIRED: Evidence is insufficient for investment approval. Validate the highest-impact assumptions before using this comparison to select an option.

Evidence confidence	Low (20.7 / 100)
Required next action	Resolve the priority evidence gaps and rerun the comparison before investment review.
Priority evidence gaps	Proposed Infrastructure.Accelerator Count, Proposed Infrastructure.Productive Utilization Pct, Workload.Peak Demand Units
Strongest sensitivities	Utilization, Growth, Price

Modeled financial comparison

Current 5-year TCO	\$24,167,789
Proposed 5-year TCO	\$9,284,563
Modeled 5-year savings	\$14,883,226
Modeled net value	\$14,972,815
Modeled ROI (net value / proposed-state TCO)	161.3%
Simple payback	1.5 months

Modeled estimates are decision-support inputs, not guarantees or financial advice.

Five-year comparison

Annual costs include visible recurring components and one-time transition costs.

Year	Current	Proposed	Delta
Year 1	\$2,614,495	\$1,285,767	\$1,328,728
Year 2	\$3,398,319	\$1,354,786	\$2,043,533
Year 3	\$4,456,480	\$1,697,711	\$2,758,769
Year 4	\$5,884,998	\$2,160,659	\$3,724,339
Year 5	\$7,813,497	\$2,785,640	\$5,027,857

Business-case measures

Three-year TCO savings	\$6,131,030
Five-year TCO savings	\$14,883,226
Five-year modeled net value	\$14,972,815
Modeled ROI (net value / proposed-state TCO)	161.3%
Break-even within five years	Yes

Report provenance

Scenario version	1
Analysis run	54bc8472-736f-4b3b-ba1e-40421d9b8240
Generated	2026-07-22T09:41:53.098766+00:00
Calculation posture	Deterministic Decimal engine; no model performs financial calculations

Scenario assumptions

All assumptions remain visible and should be replaced with approved evidence before a decision.

Workload envelope

Workload Type	inference
Model Size Billion	70.0
Monthly Requests Million	60.0
Average Demand Units	32.0
Peak Demand Units	85.0
Annual Growth Pct	35.0

Current and proposed states

Assumption	Current	Proposed
Label	CPU inference estate	GPU inference estate
Infrastructure Type	owned	owned
Accelerator Count	96	12
Compute Hourly Cost	2.4	4.8
Productive Utilization Pct	48.0	48.0
Storage Tb	36.0	36.0
Network Egress Tb Month	24.0	24.0
Power Kw	120.0	120.0
Pue	1.45	1.45
Staff Fte	2.5	2.5
Operating Hours Year	8760.0	8760.0

Transition and provenance

Migration cost	\$120,000
Implementation cost	\$65,000
Contract duration	3 year(s)
Sources	Fictional benchmark; Illustrative list pricing; Fictional Northstar peak-demand normalization; Fictional Northstar portfolio case contract; Northstar storage plan: 18 TB governed data plus working capacity

Sensitivity analysis

One-factor deterministic ranges challenge utilization, price, growth, and energy assumptions. They are scenarios, not forecasts or probability distributions.

Dimension	Case	Assumption	Proposed TCO	Net value	Modeled ROI
Utilization	Low	38.0	\$11,125,409	\$13,131,969	118.0%
Utilization	Base	48.0	\$9,284,563	\$14,972,815	161.3%
Utilization	High	58.0	\$8,078,492	\$16,178,887	200.3%
Price	Low	4.32	\$8,782,289	\$15,475,089	176.2%
Price	Base	4.8	\$9,284,563	\$14,972,815	161.3%
Price	High	5.28	\$9,786,838	\$14,470,541	147.9%
Growth	Low	30.0	\$8,623,185	\$13,585,182	157.5%
Growth	Base	35.0	\$9,284,563	\$14,972,815	161.3%
Growth	High	40.0	\$10,003,957	\$16,482,172	164.8%
Energy	Low	0.104	\$8,890,069	\$15,367,310	172.9%
Energy	Base	0.13	\$9,284,563	\$14,972,815	161.3%
Energy	High	0.156	\$9,679,058	\$14,578,321	150.6%

Interpret sensitivity as decision risk. A favorable base case with a fragile downside still requires validation.

Calculation lineage

Selected material outputs are shown below. JSON and CSV exports retain the complete analysis snapshot and lineage collection.

Output	Formula	Derived value
current.tco_3_year	sum annual total for years 1-3	10469293.72
current.tco_5_year	sum annual total for years 1-5	24167789.02
proposed.tco_3_year	sum annual total for years 1-3	4338263.64
proposed.tco_5_year	sum annual total for years 1-5	9284563.41
comparison.savings_3_year	current 3-year TCO - proposed 3-year TCO	6131030.08
comparison.savings_5_year	current 5-year TCO - proposed 5-year TCO	14883225.61
comparison.savings_3_year_pct	3-year savings / current 3-year TCO x 100	58.56
comparison.savings_5_year_pct	5-year savings / current 5-year TCO x 100	61.58
comparison.productivity_value_3_year	sum productivity value per hour x downtime hours x 12 x growth	37552.5
comparison.productivity_value_5_year	sum productivity value per hour x downtime hours x 12 x growth	89589.44
comparison.net_value_3_year	3-year TCO savings + 3-year modeled productivity value	6168582.58
comparison.net_value_5_year	5-year TCO savings + 5-year modeled productivity value	14972815.05
comparison.roi_3_year_pct	3-year net value / proposed 3-year TCO x 100	142.19
comparison.roi_5_year_pct	5-year net value / proposed 5-year TCO x 100	161.27
comparison.payback_months	transition investment / cumulative modeled monthly operating benefit	1.5
comparison.break_even_within_5_years	payback months exists and is no more than 60	true
confidence.score	source coverage weighted 80% + contract coverage weighted 20%	20.7
confidence.source_coverage_pct	source coverage weighted 80% + contract coverage weighted 20%	10.83
confidence.contract_coverage_pct	source coverage weighted 80% + contract coverage weighted 20%	60.0
confidence.level	source coverage weighted 80% + contract coverage weighted 20%	Low
executive_summary.net_value_5_year	copy or interpret the corresponding modeled result	14972815.05
executive_summary.roi_5_year_pct	copy or interpret the corresponding modeled result	161.27
executive_summary.payback_months	copy or interpret the corresponding modeled result	1.5

Output	Formula	Derived value
executive_summary.recommendation	copy or interpret the corresponding modeled result	Evidence is insufficient for investment approval. Validate the highest-impact assumptions before using this comparison to select an option.